Gonzalo Benegas

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Summary

AI for Science researcher with a PhD in Computational Biology.

Experience

Open Athena, Research Scientist – New York, NY 2025 – present
- Developing training data and evaluations to advance scientific capabilities in Marin, an openly developed LLM.
- Led the development of MarinDNA, a 1B-parameter genomic language model competitive with Evo 2 40B on variant effect prediction while scoring variants 2,300× faster.

University of California, Berkeley, Postdoctoral Researcher, EECS – Berkeley, CA 2024 – 2025
- Co-developed GPN-Star, a phylogeny-informed genomic language model achieving SOTA variant effect prediction across both rare and common variants in the human genome (Nature, 2026).
- Co-authored a review on genomic language models, featured on the cover of Trends in Genetics (2025).
- Created TraitGym, a benchmark for evaluating DNA sequence models on variant effect prediction (bioRxiv, 2025). Used as evaluation benchmark by DeepMind's AlphaGenome (Nature, 2026).

University of California, Berkeley, Graduate Student Researcher – Berkeley, CA 2018 – 2023
- Developed GPN-MSA, the first alignment-based genomic language model, achieving SOTA variant effect prediction for rare variants and scalable inference across all 9B possible SNVs in the human genome (Nature Biotechnology, 2025).
- Developed GPN, the first genomic language model to exhibit zero-shot variant effect prediction capabilities across the entire genome (PNAS, 2023).
- Developed scQuint, a variational auto-encoder for analysis of alternative splicing in single-cell RNA-seq (eLife, 2022).

Autodesk, Software Engineer – Buenos Aires, Argentina 2015 – 2016

Synthicity, Software Engineer – Buenos Aires, Argentina 2014 – 2015

Education

PhD **University of California, Berkeley**, PhD in Computational Biology – Berkeley, CA 2023
- Thesis: Computational and Machine Learning Methods for Understanding Gene Regulation and Variant Effects
- Advisor: Yun S. Song

BS+MS **University of Buenos Aires**, BS+MS in Computer Science – Buenos Aires, Argentina 2018
- GPA: 9.70/10
- Thesis: Bayesian Analysis of Transcriptional Dynamics
- Advisors: Alejo Salles and Hernan G. Garcia

Publications

Mapping cis-regulatory mutations at scale in sorghum enables modulation of gene expression 2026
Evan D. Groover, David Ding, Flora Z. Wang, **Gonzalo Benegas**, Joseph Rivera, Shahar Schwartz, Stephen Chen, Michael F. Moubarak, Viktoriya Georgieva, Peggy G. Lemaux, Brian J. Staskawicz, Krishna K. Niyogi, Yun S. Song, David F. Savage
(Nature Biotechnology)

Predicting genome-wide functional constraints with GPN-Star 2026
Chengzhong Ye†, **Gonzalo Benegas**†, Carlos Albors, Jianan Canal Li, Sebastian Prillo, Peter D. Fields, Brian Clarke, Yun S. Song
(Nature)

A DNA language model based on multispecies alignment predicts the effects of genome-wide variants 2025
Gonzalo Benegas, Carlos Albors, Alan J. Aw, Chengzhong Ye, Yun S. Song
(Nature Biotechnology, 43, 1960–1965)

Genomic language models: opportunities and challenges 2025
Gonzalo Benegas†, Chengzhong Ye†, Carlos Albors†, Jianan Canal Li†, Yun S. Song
(Trends in Genetics, 41(4), 286–302)

Benchmarking DNA Sequence Models for Causal Regulatory Variant Prediction in Human Genetics 2025
Gonzalo Benegas, Gökcen Eraslan, Yun S. Song
(bioRxiv)

A Phylogenetic Approach to Genomic Language Modeling 2025
Carlos Albors, Jianan Canal Li, **Gonzalo Benegas**, Chengzhong Ye, Yun S. Song
(RECOMB 2025, pp. 99–117)

DNA language models are powerful predictors of genome-wide variant effects 2023
Gonzalo Benegas, Sanjit Singh Batra, Yun S. Song
(Proceedings of the National Academy of Sciences, 120(44), e2311219120)

choros: correction of sequence-based biases for accurate quantification of ribosome profiling data 2023
Amanda Mok, Robert Tunney, **Gonzalo Benegas**, Edward W. J. Wallace, Liana F. Lareau
(bioRxiv)

Robust and annotation-free analysis of alternative splicing across diverse cell types in mice 2022
Gonzalo Benegas, Jonathan Fischer, Yun S. Song
(eLife, 11, e73520)

Presentations

Invited talk, Kipoi Seminar – Virtual 2025

Invited talk, Google Research Genomics Seminar – Virtual 2025

Invited talk, Variant Effects Seminar Series – Virtual 2024

Poster, Machine Learning in Computational Biology (MLCB) – University of Washington 2024

Talk, AI≡Science workshop, Simons Institute for the Theory of Computing 2024

Invited talk, Debora Marks Lab, Harvard 2023

Invited talk, iGEM at Berkeley Journal Club – UC Berkeley 2023

Talk, Machine Learning in Computational Biology (MLCB) – Virtual 2022

Invited talk, Moi Exposito-Alonso Lab, Stanford 2022

Invited talk, Luca Pinello Lab, Harvard 2022

Talk, Center for Computational Biology Annual Retreat – UC Berkeley 2021

Poster, CZI Seed Networks 2020 Annual Meeting – Chan Zuckerberg Initiative 2020

Teaching

UC Berkeley, Graduate Student Instructor 2021
BIOENG C131: Introduction to Computational Molecular and Cell Biology

UC Berkeley, Graduate Student Instructor 2019
CMPBIO 293: Doctoral Seminar in Computational Biology

Awards

Reviewers' Choice Poster, ASHG Annual Meeting – Washington, DC 2023

Best poster, runner-up, Center for Computational Biology Annual Retreat – UC Berkeley 2022

Secondary School Exchange Scholar, English-Speaking Union 2011

Second Top Score in Argentina, Cambridge AICE Diploma 2010

Top Score in Argentina, Cambridge AS Level Mathematics 2010

Top Score in Argentina, Cambridge AS Level Biology 2010

Honorable Mention, Argentine National Mathematical Olympiad 2010

Champion, Buenos Aires Province Mathematical Olympiad 2010

Top Score in Argentina, Cambridge IGCSE Chemistry 2009

Top Score in Argentina, Cambridge IGCSE Biology 2009

Champion, Buenos Aires Province Mathematical Olympiad 2009

Service

Reviewer, Nature Methods 2025

Reviewer, Genome Biology 2025

Reviewer, Theoretical and Applied Genetics 2025

Reviewer, Cal Undergraduate BioE Symposium, UC Berkeley 2023

Co-organizer, Center for Computational Biology Student Seminar, UC Berkeley 2021 – 2023

Co-organizer, Center for Computational Biology Seminar, UC Berkeley 2018 – 2019

Questioning Panel Member, National Public Speaking Competition, English-Speaking Union 2014